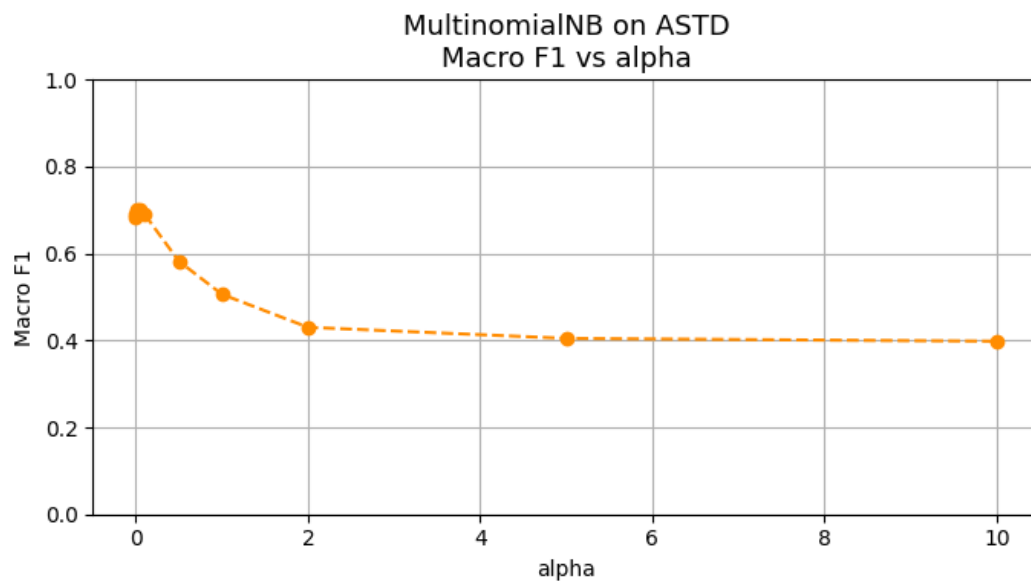
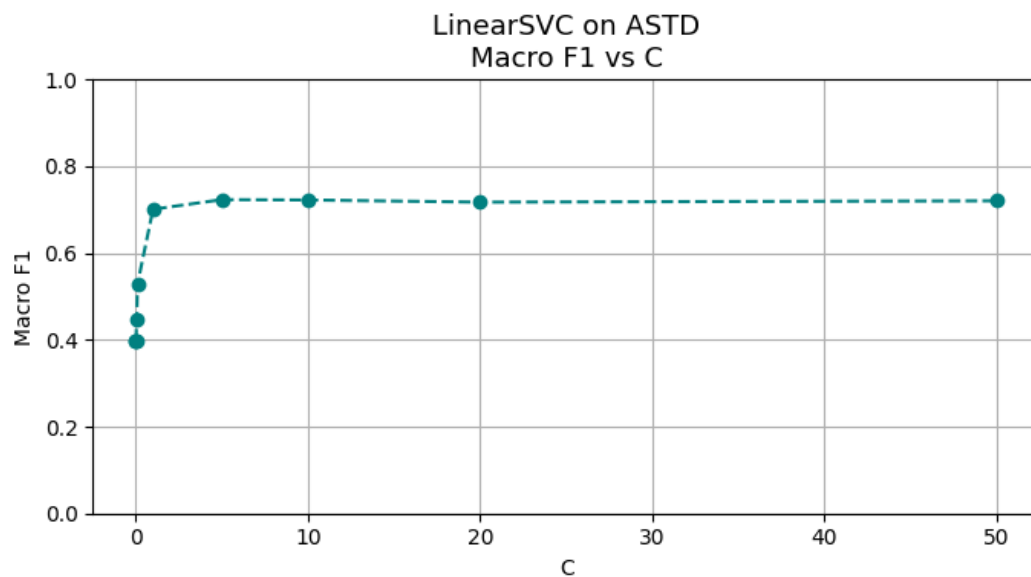


Analyse de l'effet des hyperparamètres pour chaque algorithme

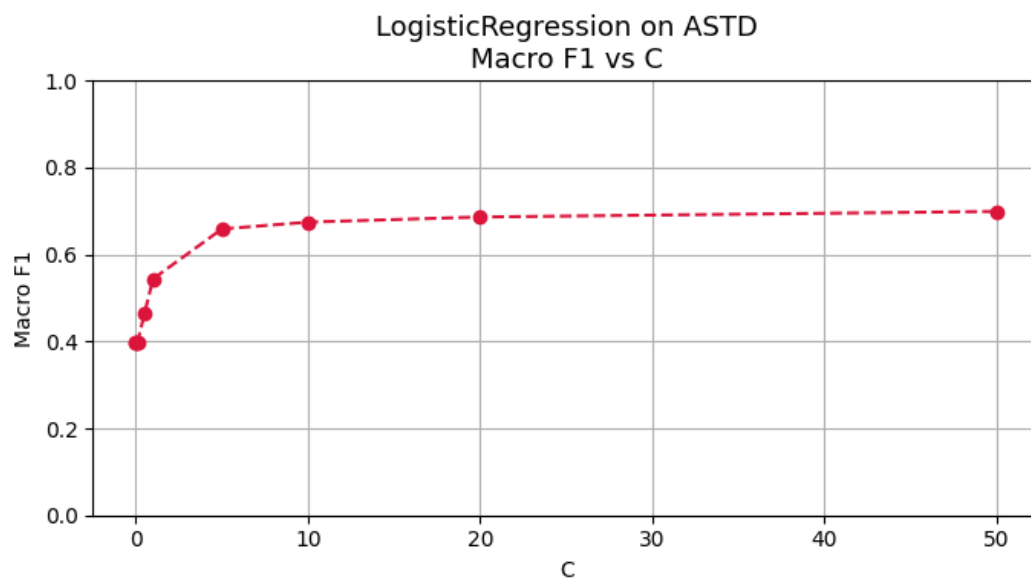
MultinomialNB (alpha)



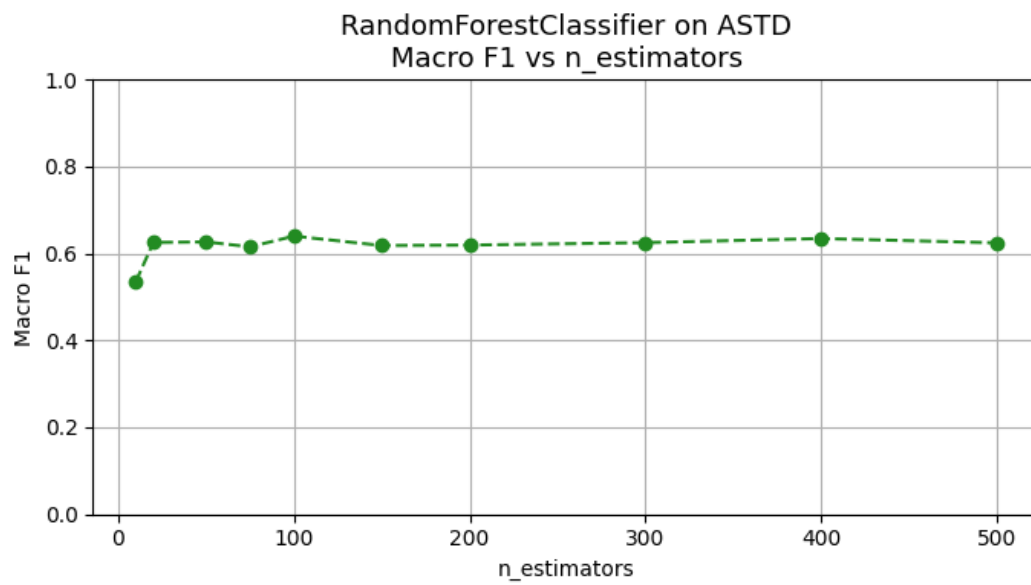
LinearSVC (C)



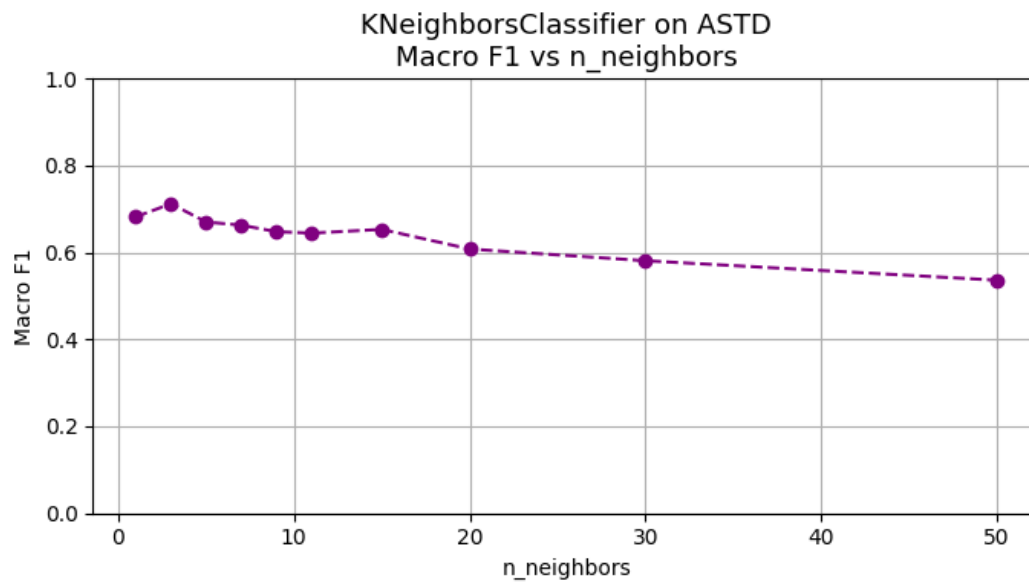
LogisticRegression (C)



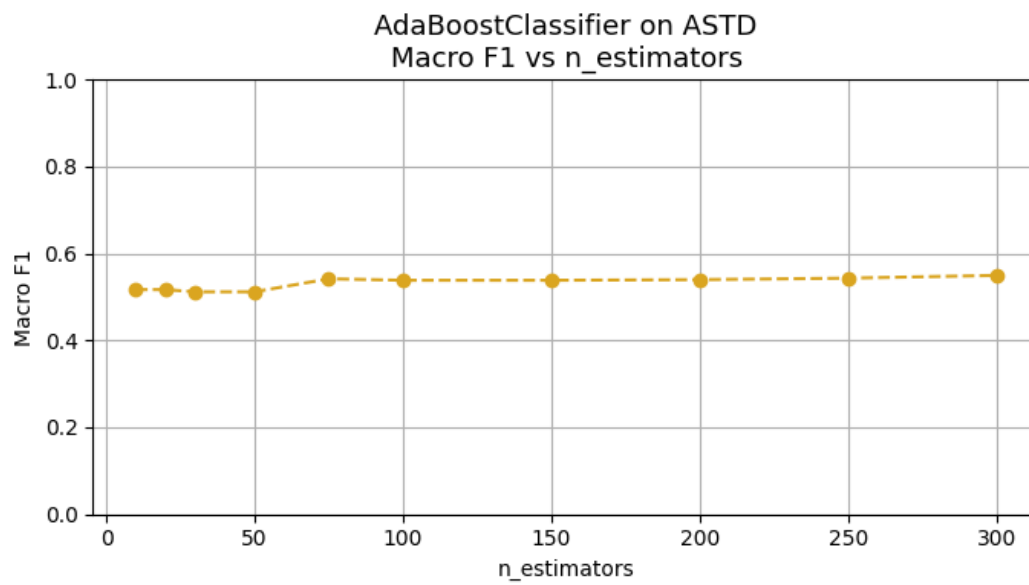
RandomForestClassifier (n_estimators)



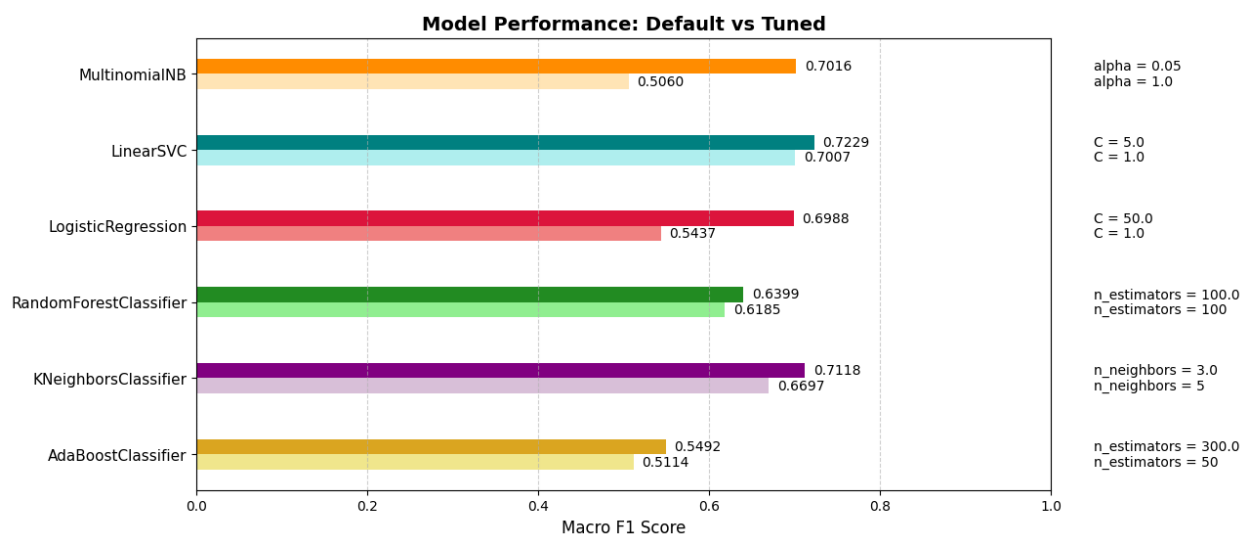
KNeighborsClassifier (n_neighbors)



AdaBoostClassifier (n_estimators)



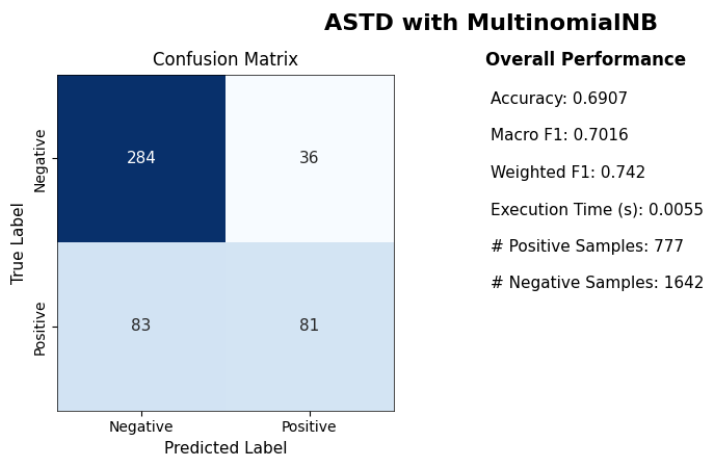
Comparaison des performances: Hyperparamètres par défaut vs Optimisés



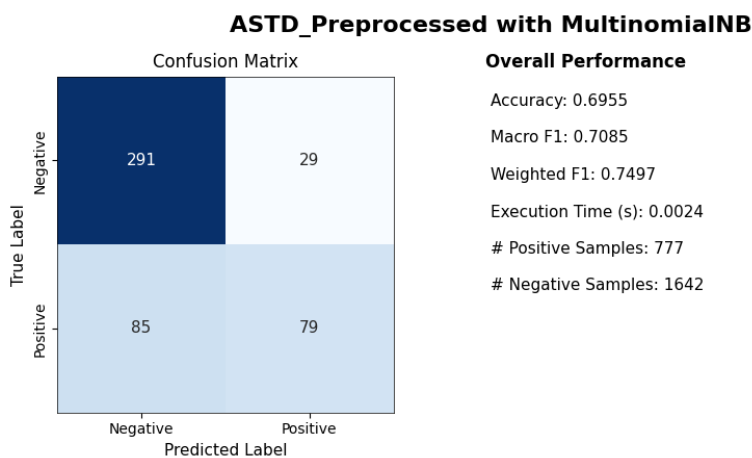
Évaluation comparative sur différentes versions du dataset

Multinomial Naïve Bayes

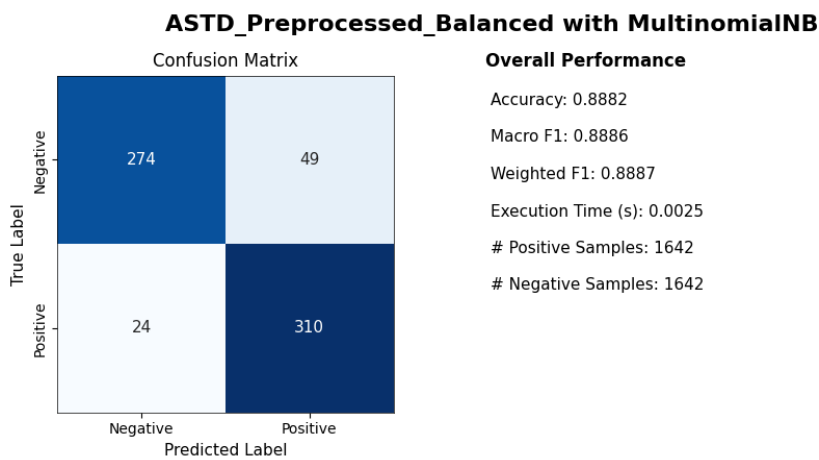
ASTD (Brut)



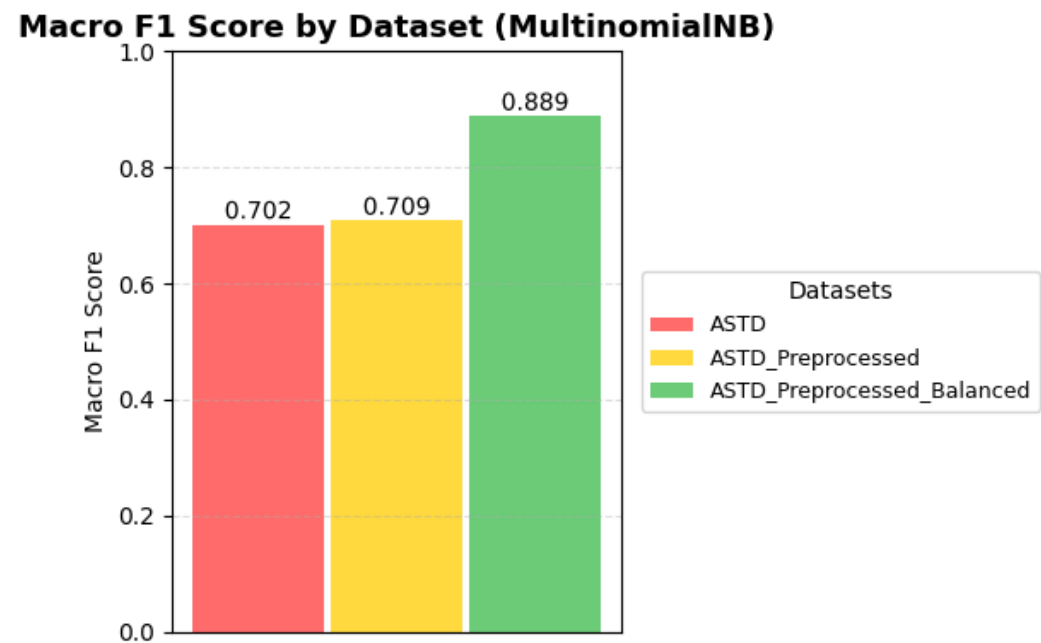
ASTD (Prétraité)



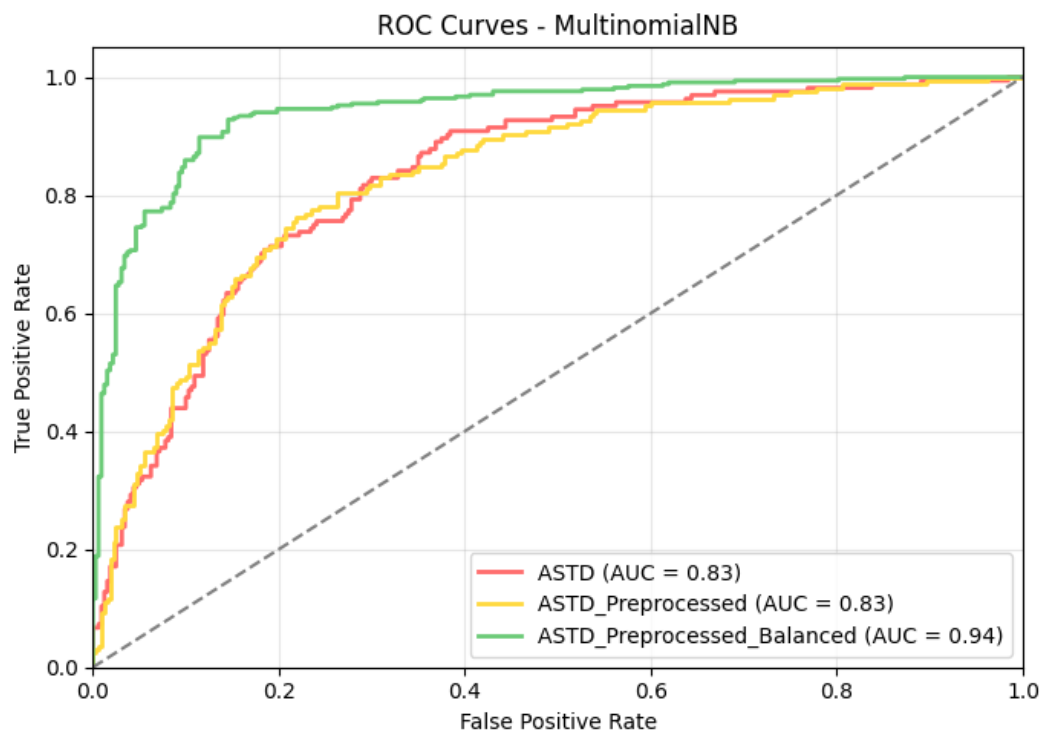
ASTD (Prétraité + Équilibré)



Comparaison: Macro F1



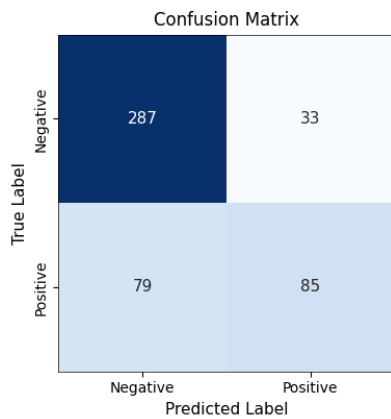
Comparison: ROC



Logistic Regression

ASTD (Brut)

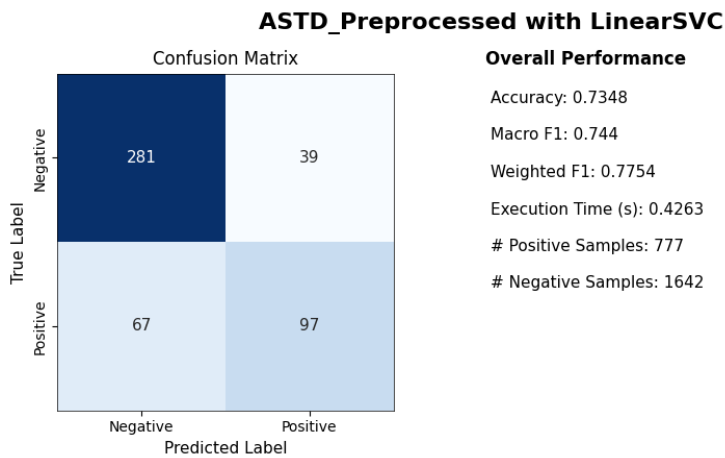
ASTD with LinearSVC



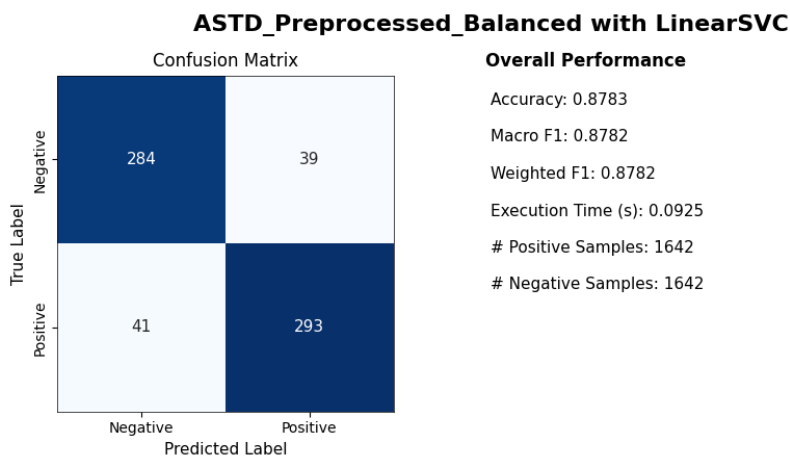
Overall Performance

Accuracy: 0.7076
Macro F1: 0.7198
Weighted F1: 0.7575
Execution Time (s): 0.8845
Positive Samples: 777
Negative Samples: 1642

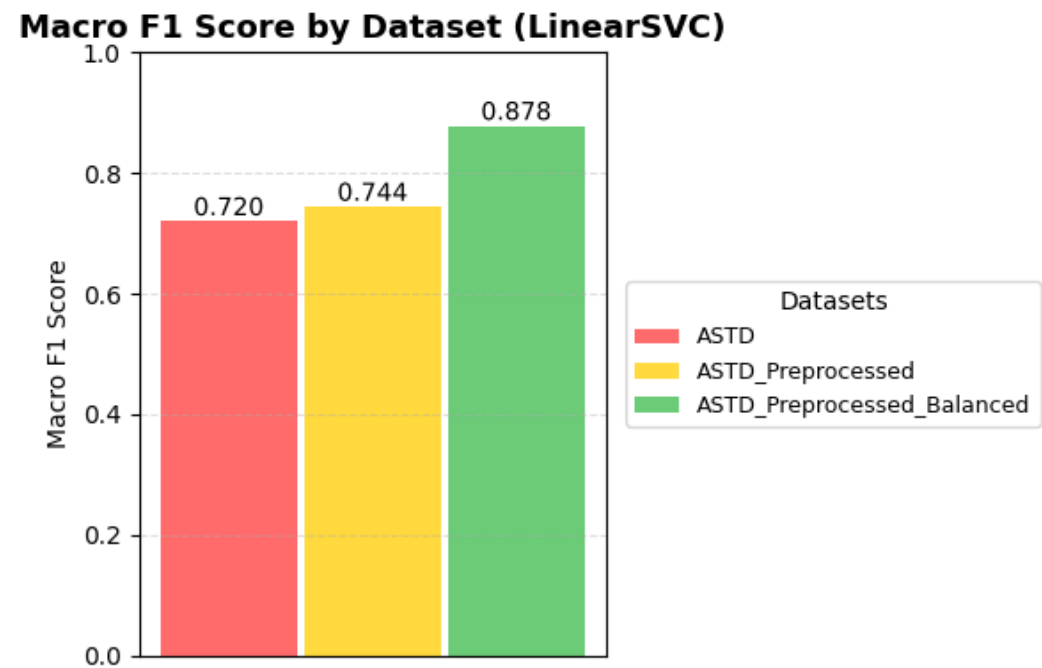
ASTD (Prétraité)



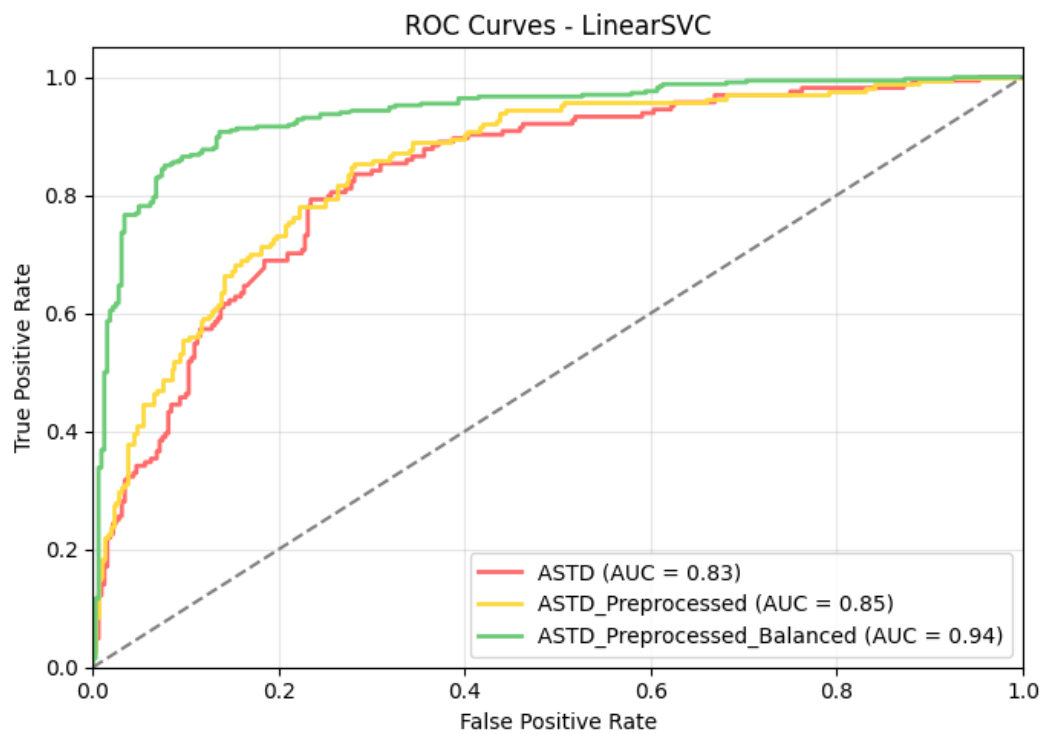
ASTD (Prétraité + Équilibré)



Comparaison: Macro F1



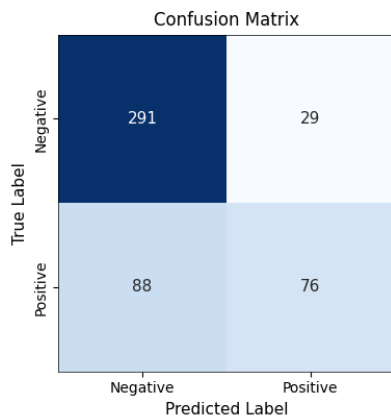
Comparaison: ROC



SVM-Linear

ASTD (Brut)

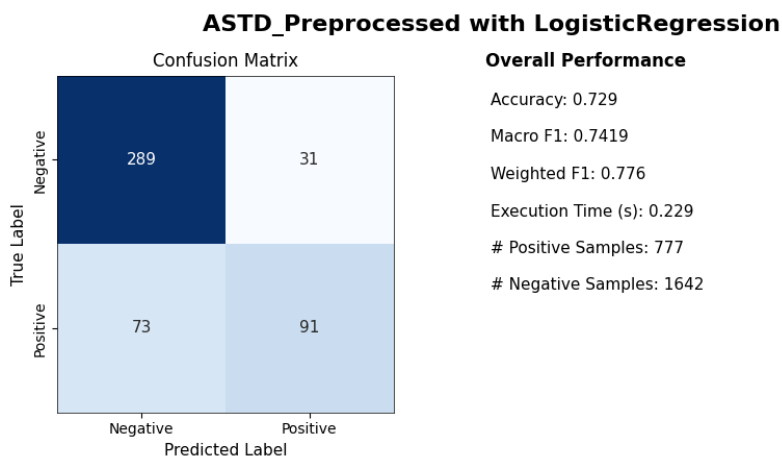
ASTD with LogisticRegression



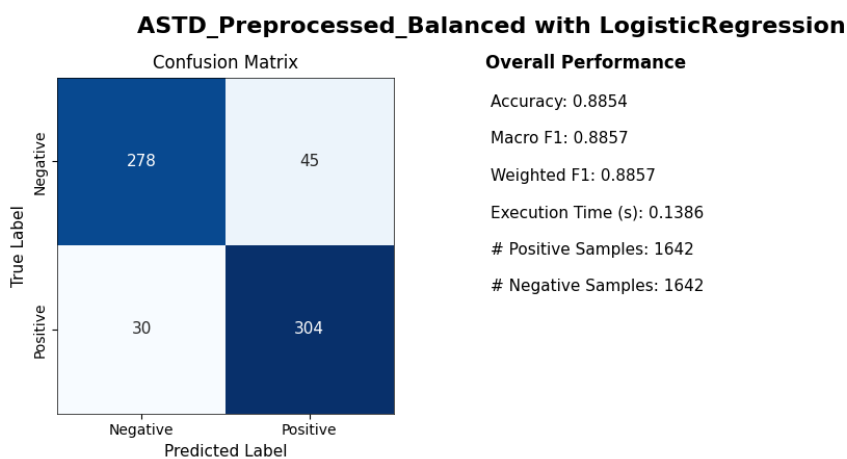
Overall Performance

Accuracy: 0.6864
Macro F1: 0.6988
Weighted F1: 0.742
Execution Time (s): 0.94
Positive Samples: 777
Negative Samples: 1642

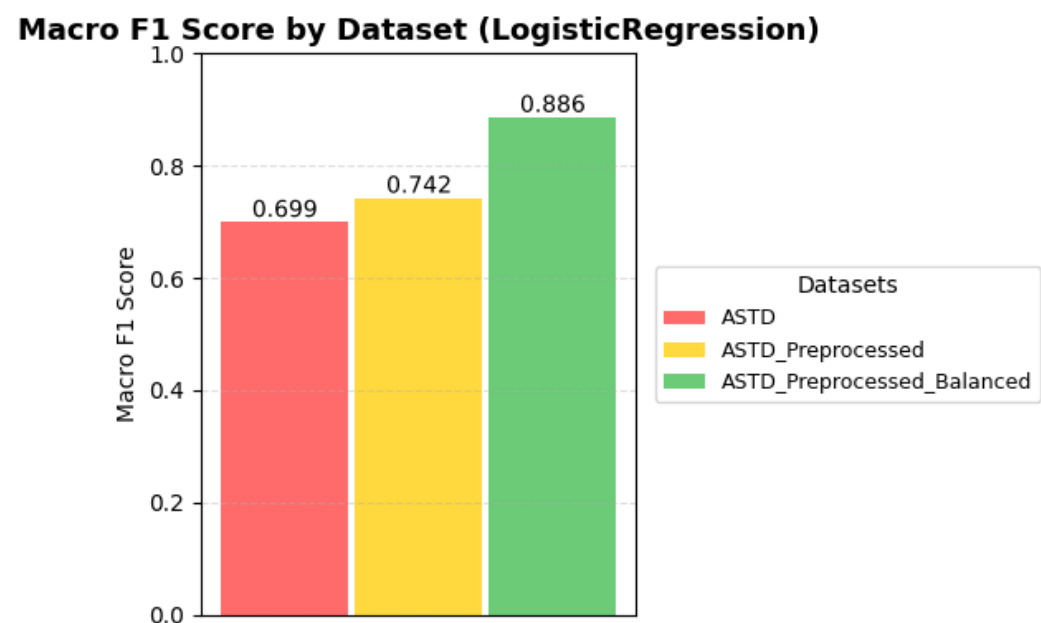
ASTD (Prétraité)



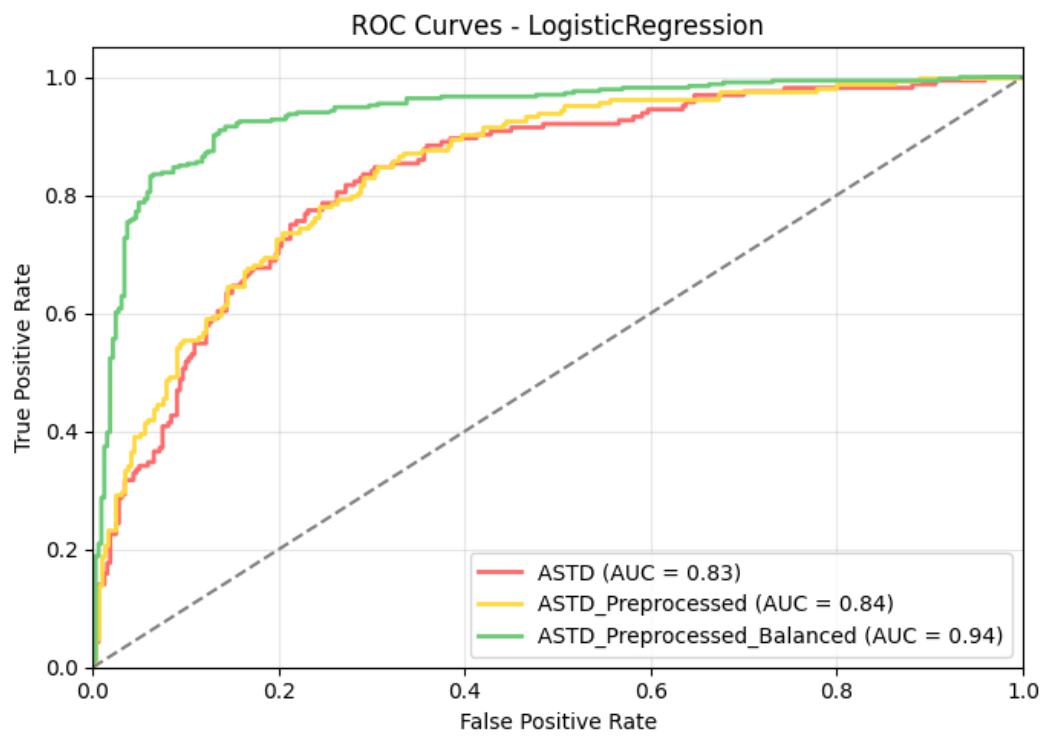
ASTD (Prétraité + Équilibré)



Comparaison: Macro F1



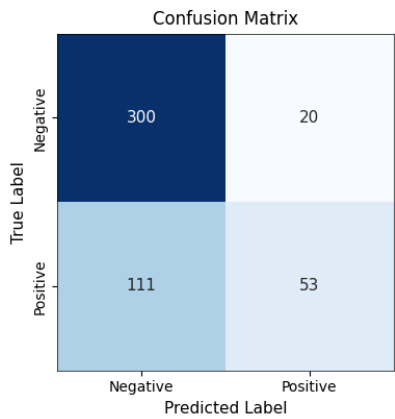
Comparison: ROC



Random Forest

ASTD (Brut)

ASTD with RandomForestClassifier

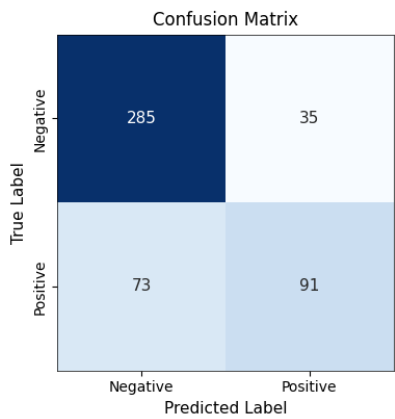


Overall Performance

Accuracy: 0.6303
Macro F1: 0.634
Weighted F1: 0.6942
Execution Time (s): 2.2173
Positive Samples: 777
Negative Samples: 1642

ASTD (Prétraité)

ASTD_Preprocessed with RandomForestClassifier

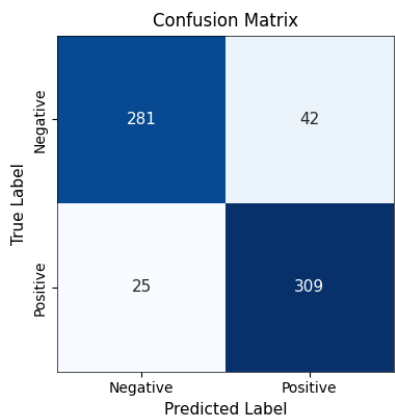


Overall Performance

Accuracy: 0.7228
Macro F1: 0.7341
Weighted F1: 0.7685
Execution Time (s): 0.7999
Positive Samples: 777
Negative Samples: 1642

ASTD (Prétraité + Équilibré)

ASTD_Preprocessed_Balanced with RandomForestClassifier

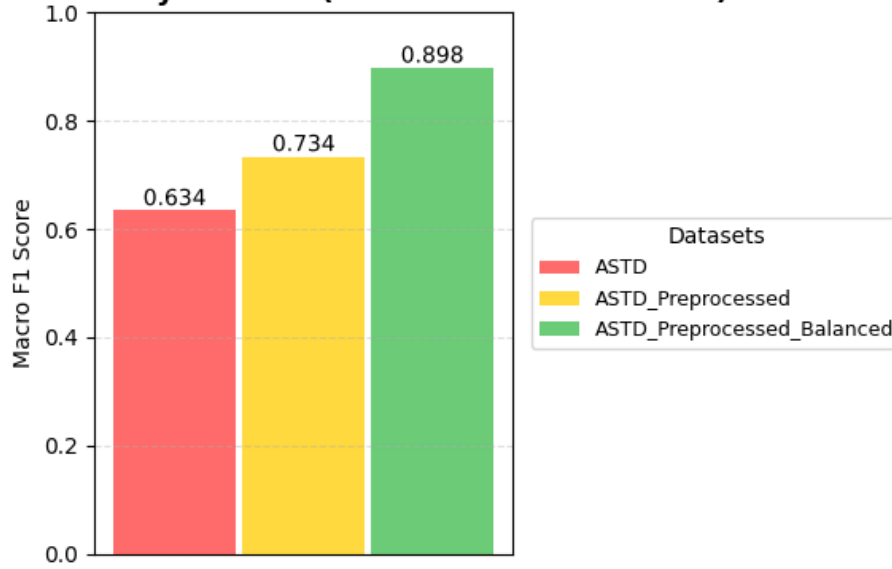


Overall Performance

Accuracy: 0.8976
Macro F1: 0.8978
Weighted F1: 0.8979
Execution Time (s): 1.8048
Positive Samples: 1642
Negative Samples: 1642

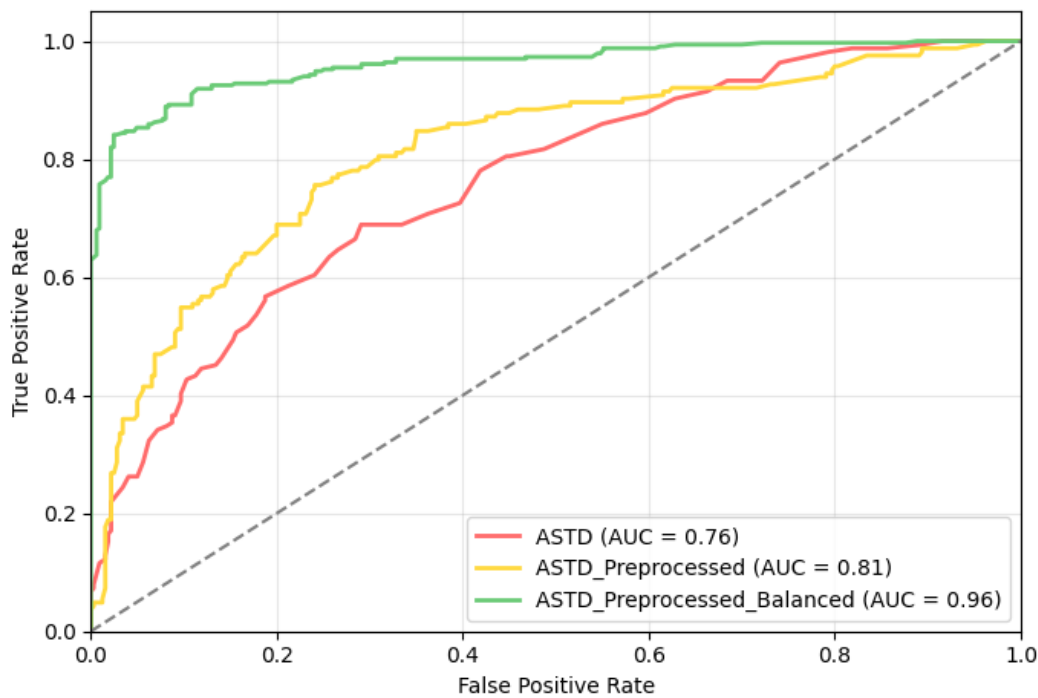
Comparison: Macro F1

Macro F1 Score by Dataset (RandomForestClassifier)



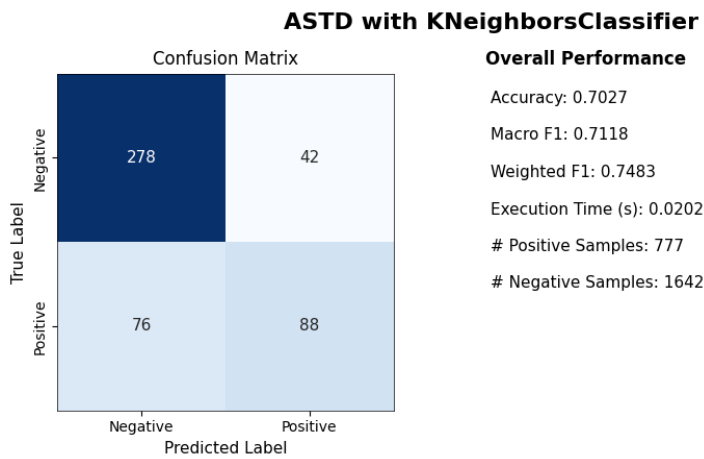
Comparison: ROC

ROC Curves - RandomForestClassifier

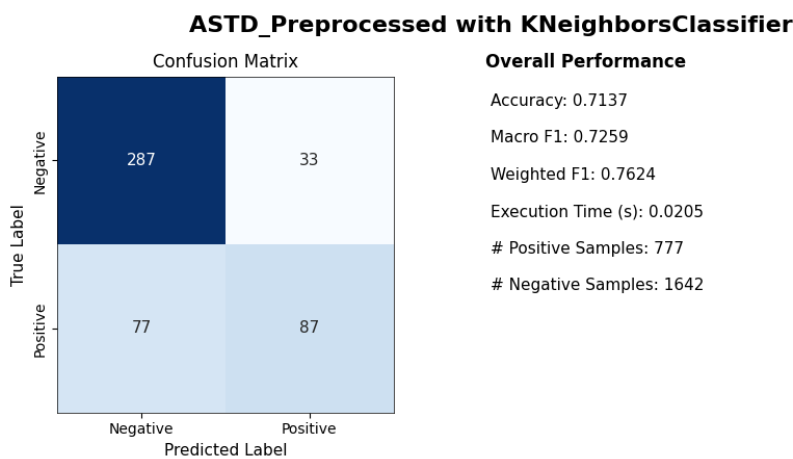


K-Nearest Neighbors

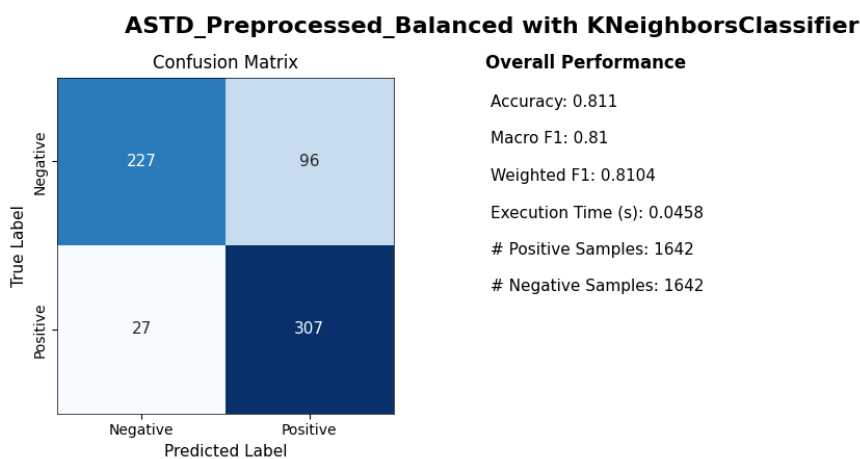
ASTD (Brut)



ASTD (Prétraité)

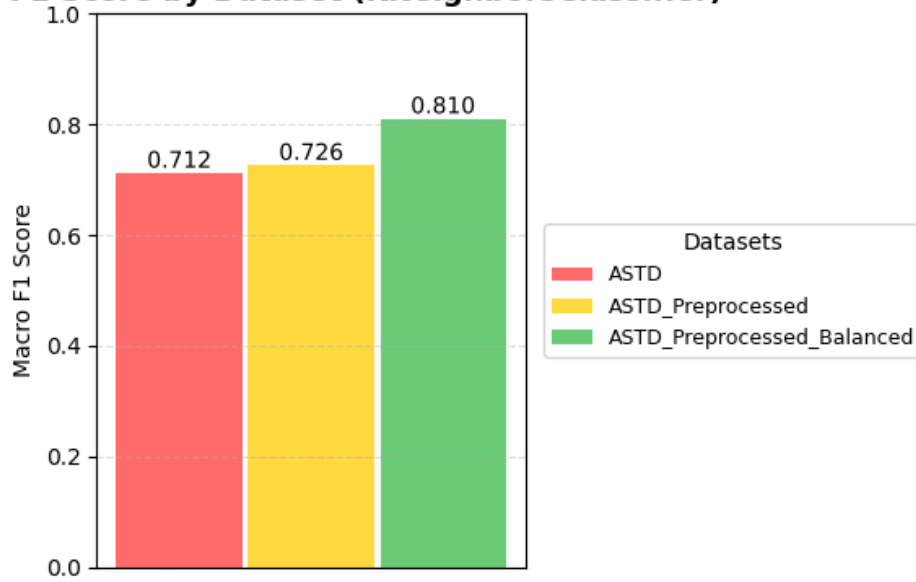


ASTD (Prétraité + Équilibré)



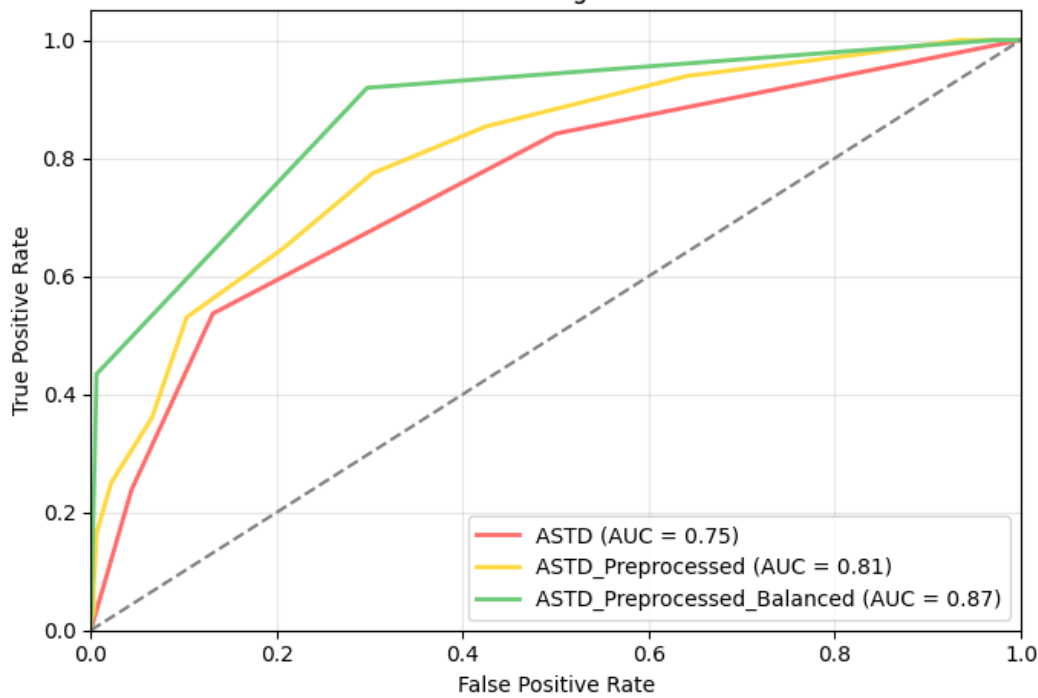
Comparison: Macro F1

Macro F1 Score by Dataset (KNeighborsClassifier)



Comparison: ROC

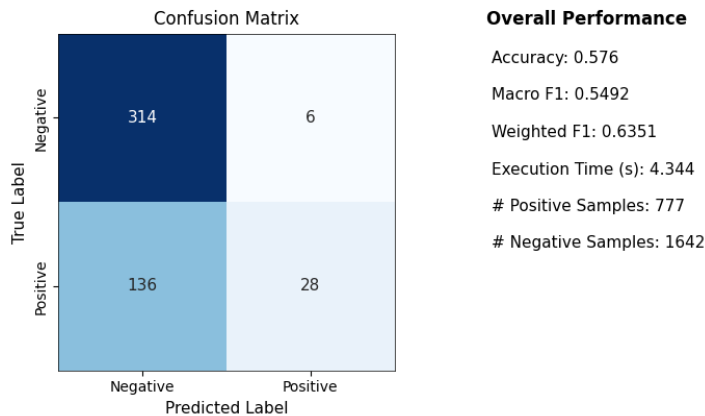
ROC Curves - KNeighborsClassifier



AdaBoost Classifier

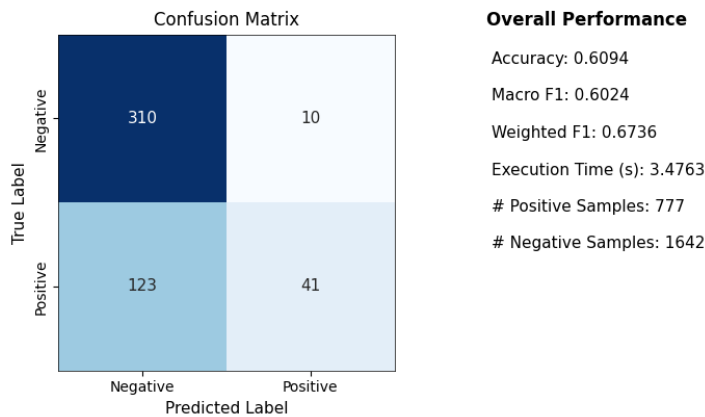
ASTD (Brut)

ASTD with AdaBoostClassifier



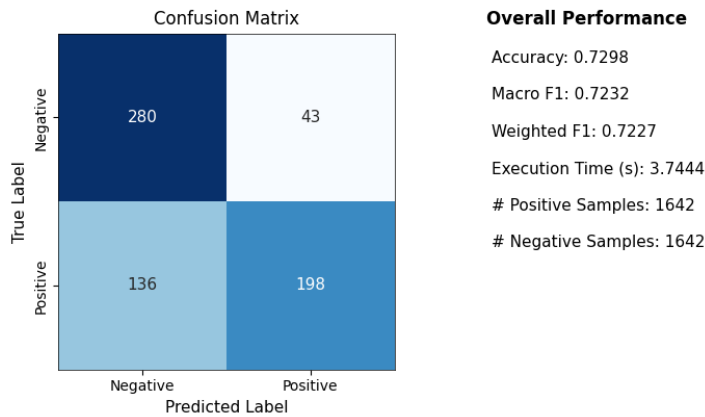
ASTD (Prétraité)

ASTD_Preprocessed with AdaBoostClassifier



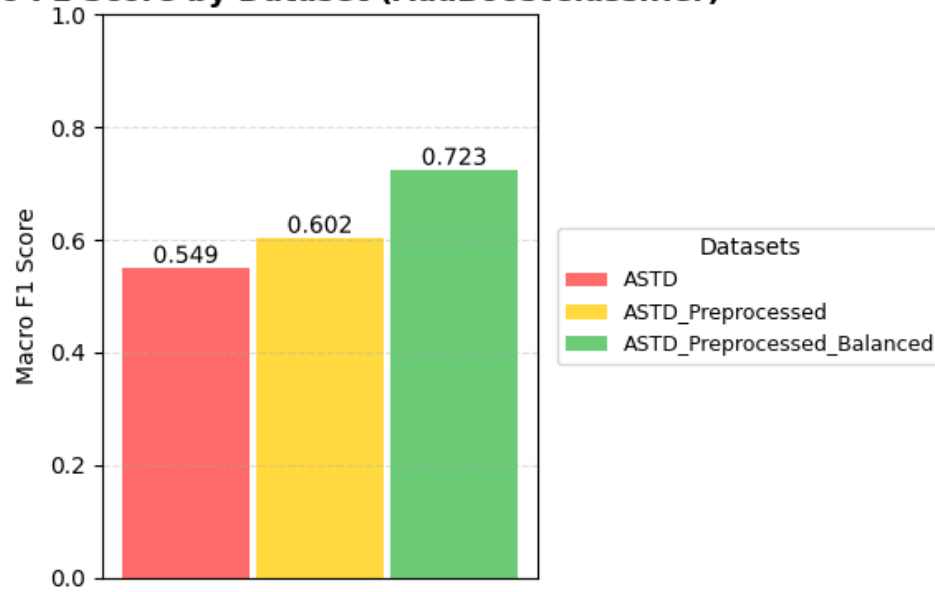
ASTD (Prétraité + Équilibré)

ASTD_Preprocessed_Balanced with AdaBoostClassifier



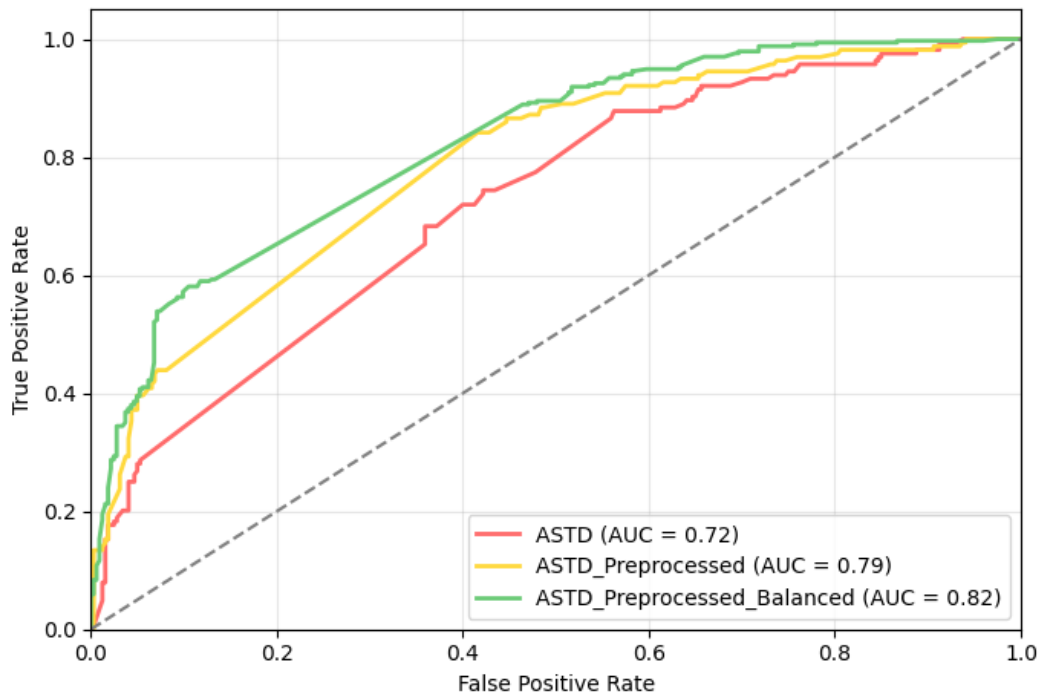
Comparison: Macro F1

Macro F1 Score by Dataset (AdaBoostClassifier)



Comparison: ROC

ROC Curves - AdaBoostClassifier



Global Macro F1 Comparison Across All Models

